Model 1 Writing Brief

Coupled vegetation–water–sand dynamics (Task 1)

Source of every number: `Results/v7/model1/` (run of `Code/v7_model1.py`)

2026-11-11

This brief collects what you need to write the Model 1 section of the paper: the goal, how the model is built, every formula with its meaning, parameter values, the results with numbers, and what the results do and do not allow you to claim. All values are copied from the result tables; figure files are in `Figures/M1/`. Units: V is in percentage points of a cover proxy, M in mm, irrigation in mm/yr on the treated area (1 mm = 10 m^3/ha).

Contents

1 Goal and what we achieved **1**

2 Model construction **2**

3 Formulas and what they mean **2**
- 3.1 Vegetation update (yearly) . . . 2
- 3.2 Effective soil-water storage (daily sub-steps) . . . 3
- 3.3 Helper relations . . . 3
- 3.4 Where wind enters . . . 3
- 3.5 Long-run state and the closed form . . . 4
- 3.6 Calibration and checks . . . 4

4 Algorithms **5**

5 Parameters **7**

6 Results **7**
- 6.1 Calibration and model selection . . . 7
- 6.2 Irrigation–cover curve (long-run state) . . . 8
- 6.3 Scenario paths 2026–2050 . . . 8
- 6.4 Sensitivity . . . 9

7 What we can claim and what we cannot **9**

8 Figures ready for the paper **10**

9 Suggested outline for the Model 1 section **11**

1 Goal and what we achieved

Task 1 asks for a coupled dynamic model of vegetation cover and effective soil moisture on a representative section of the desert margin, using real data, and for the factors that drive them.

Requirement	How Model 1 meets it	Evidence
Representative section (width tens of m to km, length tens to hundreds of km)	Yutian–Minfeng southern margin of the Taklimakan, centre line through windows CON1–CON4, length 127.95 km, assumed width 1 km	`section_definition.json`
Coupled *dynamic* model	Two-state difference system: yearly cover V_t and within-year storage M_j, solved together every year	Eqs. (1)–(5)
State variables V and M	V: growing-season NDVI cover proxy; M: plant-available water above wilting point (mm)	Section 3
Use data; find influencing factors	MODIS NDVI 2015–2025 at 7 stations, daily wind, monthly P, ET_0, T; influence of irrigation, rain, demand, efficiency, wind speed quantified	Tables 4, 6
Water–vegetation–sand feedback	Water → growth ($\overline{W}$); cover → water demand; wind → evaporative demand → $\overline{W}$; cover → sand exposure S	Fig. 1

Table 1: Task 1 requirements and how they are met.

Main results in one paragraph. Pooled over 7 oasis stations, the intrinsic growth rate is $r = 0.547$ /yr and the decay rate $d = 0.107$ /yr; cover would halve in about 6.1 years without water. Under 2015–2022 climate (52 mm/yr rain against 1,686 mm/yr reference evapotranspiration), the long-run cover proxy rises from 1.8% with no irrigation to 17.9% at 300 mm/yr and 24.1% at 600 mm/yr, with sharply falling returns, and saturates at 30.6% above about 1,500 mm/yr. Stopping irrigation after 10 years drops cover from 21.5% to 5.1% within 15 years. The one-year-ahead forecast does **not** beat "next year equals this year" on held-out years (RMSE 0.0635 vs 0.0536), so the model is a scenario tool, not a forecaster.

2 Model construction

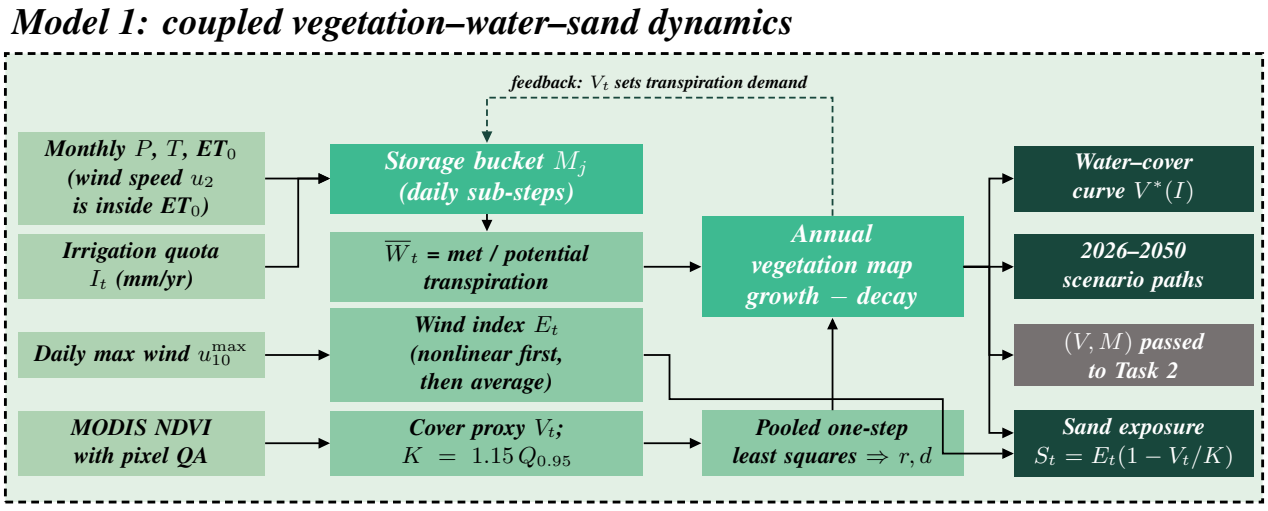


Figure 1: Structure of Model 1. Inputs on the left; the storage bucket runs on daily sub-steps inside each year; the vegetation map updates once per year; outputs on the right.

Two states, two time scales.

- V_t (yearly): mean NDVI-derived cover proxy of the June–September growing season of year t, with end-members NDVI 0.05 (bare) and 0.85 (full): $V = 100\,\mathrm{clip}\big((\mathrm{NDVI} - 0.05)/0.80, 0, 1\big)$.

- M_j (daily sub-steps j within each year, carried across years): water above wilting point in the root zone, $0 \le M_j \le C$.

Order of computation in each year. (1) Hold V_t fixed; (2) run the storage bucket over every day of the year with that year's rain, ET_0 and irrigation; (3) sum transpiration into the met-demand ratio $\overline{W}_t$; (4) update V_{t+1} with $\overline{W}_t$; (5) carry the year-end storage to next year; (6) report the sand-exposure index S_t.

Two processes in the vegetation equation. Logistic growth limited by water and a local ceiling K; first-order natural decay. There is no wind term in the vegetation equation (changed 2026-11-11, Section 3.4).

Four feedback links. water → vegetation (growth is multiplied by $\overline{W}$); vegetation → water (more cover means more transpiration demand, so lower $\overline{W}$); wind → water → vegetation (wind speed raises ET_0, hence demand, hence lowers $\overline{W}$); vegetation → sand (exposure index $S = E(1 - V/K)$, an output).

Writing note: Present Fig. 1 first, then the vegetation equation, then the bucket. Stress that $\overline{W}$ is a transpiration ratio, not soil moisture; M is the soil-water state.

3 Formulas and what they mean

3.1 Vegetation update (yearly)

$$V_{t+1} = V_t + \underbrace{rV_t\left(1 - \frac{V_t}{K}\right)\overline{W}_t}_{\text{growth}} - \underbrace{dV_t}_{\text{natural decay}} \tag{1}$$

- **Growth.** rV is exponential growth; $(1 - V/K)$ stops growth at the local ceiling K; $\overline{W}_t \in [0, 1]$ scales growth by how much of the plants' water demand was met. Temperature is already inside r (one yearly growth season), so no extra temperature factor is multiplied here.
- **Decay.** A fixed share d of cover dies each year. With no water, $V_{t+1} = (1 - d)V_t$, so the half-life is $\ln(1/2)/\ln(1 - d)$ years.
- **No explicit wind term in the calibrated equation (rev. 2026-11-12).** The wind term $kE_{\mathrm{eff}}V$ is *kept in the model form* so that the third channel stays visible and Task 3 keeps a physical entry point, but k is **not estimated**: the baseline is $k = 0$, $\eta = 1$. Two reasons. (i) *Structure:* at $E = 1, \eta = 1$ the term equals $k(1 - V/K)V$, the same shape as growth $r(1 - V/K)V\overline{W}$, so with $\overline{W}$ and E both close to constant only $r - k$ is identified. (ii) *Evidence:* the pooled fit gives $\hat{k} = 0.25 \pm 0.43$ ($p = 0.57$), bootstrap 95% $[-0.44, 0.89]$, AIC worse by 1.6, and within stations the wind index is unrelated to next year's cover change ($r = 0.03$, $p = 0.81$, $n = 70$). Wind still acts through the water demand and the sand output (Section 3.4).
- $V = 0$ stays at 0: water alone cannot create plants; new stands need a planted initial cover.

3.2 Effective soil-water storage (daily sub-steps)

Monthly rain P, ET_0 and the month's share of irrigation are spread evenly over the days of that month. Potential transpiration on day j:

$$D_j = c\ \min(1, 2f_{c,t})\ f(T_j)\ \mathrm{ET}_{0,j}, \qquad f_{c,t} = \mathrm{clip}(V_t/V_{\mathrm{full}}, 0, 1). \tag{2}$$

Demand grows with cover until V reaches half of V_{full}, and follows the season through $f(T)$. Yearly irrigation I_t is split over months in proportion to $f(T)\text{ET}_0$ (watering follows demand).

$$B_j = M_j + \alpha_P P_j + \eta_w I_j + G_j \qquad \text{water available today} \tag{3}$$

$$Q_j = \max(B_j - C, 0) \qquad \text{overflow to deep drainage}$$

$$K_{s,j} = \min\{1,\ (B_j - Q_j)/[(1-p)C]\} \qquad \text{FAO-56 stress coefficient}$$

$$T_{a,j} = \min(B_j - Q_j,\ K_{s,j} D_j) \qquad \text{actual transpiration}$$

$$M_{j+1} = B_j - Q_j - T_{a,j} \qquad \text{storage carried to next day} \tag{4}$$

$$\overline{W}_t = \textstyle\sum_{j\in t} T_{a,j} \,/\, \sum_{j\in t} D_j \qquad \text{share of demand met in year } t \tag{5}$$

Every step conserves water exactly: $M_{j+1} - M_j = \alpha_P P_j + \eta_w I_j + G_j - T_{a,j} - Q_j$.

- α_P and η_w are the shares of rain and of gross irrigation that reach the roots; interception, soil evaporation and conveyance losses are inside them.
- Plants are unstressed while storage is above $(1-p)C$; below that, transpiration falls linearly.
- When storage is full, extra water drains (Q_j) and does not raise $\overline{W}$.

3.3 Helper relations

$$f(T) = \min\{1, \max\{0, (T-5)/20\}\} \qquad 0 \text{ below } 5\,^\circ\text{C},\ 1 \text{ above } 25\,^\circ\text{C} \tag{6}$$

$$q_i = u_i^2 \max(u_i - u_T, 0),\ \ u_T = 5 \text{ m/s} \qquad \text{daily sand-drift proxy from daily max wind} \tag{7}$$

$$E_t = \frac{\frac{1}{n_t}\sum_{i\in t} q_i/\bar{q}}{\text{training-period mean of the same}} \qquad \text{wind index, 1 = training mean} \tag{8}$$

$$K = 1.15 \times Q_{0.95}(V_{\text{train}}) \qquad \text{local ceiling from the 95th percentile} \tag{9}$$

The wind proxy is computed on each day and averaged afterwards; averaging the wind first would hide most gusts above the 5 m/s threshold and make E almost zero.

3.4 Where wind enters

Wind is in the model twice, and neither place is the vegetation equation.

1. **Wind → water → cover.** ET_0 in the demand D_j is FAO-56 Penman–Monteith:

$$\text{ET}_0 = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\dfrac{900}{T+273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)}, \tag{10}$$

   with u_2 the 2 m wind speed. On the section the wind-driven second term is 61% of annual ET_0. More wind ⇒ more demand ⇒ lower $\overline{W}$ ⇒ less cover. To quantify it we scale u_2, recompute ET_0 month by month and multiply the model's ET_0 forcing by the ratio (our Penman–Monteith from POWER inputs matches the provider ET_0: monthly $r = 0.990$, 1,683 vs 1,686 mm/yr).

2. **Cover → sand.** Output only:

$$S_t = E_t\Big(1 - \frac{V_t}{K}\Big), \tag{11}$$

   wind index times the unsheltered share of the surface ($S = 1$: bare sand under average wind). Scenario tables use $E = 1$.

Why the old term $-kE_t(1-\eta V_t/K)V_t$ was removed. (i) *Data.* Nine wind indicators, each with and without the shelter factor (18 fits), were added to the pooled (r, d) fit (`wind_term_test.csv`). With the index E_t: $k = 0.25 \pm 0.43$, $p = 0.57$, bootstrap 95% $[-0.44, 0.89]$, AIC worse by 1.6, hold-out RMSE 0.0657 against 0.0635 without wind. One of 18 fits has $p < 0.05$ (days with wind ≥ 8 m/s, $p = 0.036$), which is what chance gives, and it raises the hold-out RMSE to 0.077. Within stations, the wind index is unrelated to next

year's cover change ($r = 0.03$, $p = 0.81$, $n = 70$). (ii) *Structure.* At average wind ($E = 1$, $\eta = 1$) the term is $k(1 - V/K)V$, the same shape as growth: only $r - k$ can be estimated, so a constant-wind term adds a parameter and no information. (iii) *Scope.* The stations are irrigated oases; the test cannot rule out damage to young stands on open dunes or in single storms. Those belong to the shocks of Task 3.

Detectable-effect bound (added 2026-11-12). With SE = 0.43, the minimum detectable effect at 80% power ($\alpha = 0.05$) is 2.80 SE ≈ 1.21, and the one-sided 95% upper bound is $k \leq 0.89$. Converted at $\bar{V} = K/2$ this is 8.5 pp/yr of cover loss, i.e. 4–8 times the observed inter-annual variability. The data therefore *exclude strong* wind erosion but *cannot exclude* weak effects ($k < 0.5$): write "not detected", never "no effect". Task 3 re-uses k as a scenario lever anchored at this bound ($k_0 \leq 0.89$, amplified by $\kappa_t \in \{2, 5\}$ in shock years).

Writing note: Say that wind is in the model through evaporative demand and through the sand-exposure output, and that a direct damage term was tested and not detected (95% upper bound $k \leq 0.89$). Do not say that wind has no effect on plants.

3.5 Long-run state and the closed form

With $x = V/K$ and $\overline{W}$ held fixed, the non-zero steady state is

$$x^* = 1 - \frac{d}{r\overline{W}}, \qquad \text{exists if } r\overline{W} > d. \tag{12}$$

Because $\overline{W}$ itself depends on V, the real long-run state is the fixed point of the yearly map $(V, M) \mapsto (V', M')$, found by iterating from a low and a high start until both agree (residual $< 10^{-7}$). With ample water ($\overline{W} = 1$): $x^* = 1 - d/r = 0.804$, i.e. $V^* = 30.6\%$ at $K = 38.08\%$, which is exactly the saturation value found numerically.

3.6 Calibration and checks

$$\min_{r,d} \frac{1}{N} \sum_{(s,t)\in\text{train}} \left(\tilde{x}_{s,t}\left[1 + r(1 - \tilde{x}_{s,t}) - d\right] - \tilde{x}_{s,t+1}\right)^2 \tag{13}$$

One-step transitions at 7 irrigated oasis stations, pooled, assuming $\overline{W} = 1$ there; training = target year ≤ 2022 (49 transitions), hold-out = 2023–2025 (21 transitions). The residual is linear in (r, d), so the least-squares solution is unique. Uncertainty: local covariance, leave-one-station-out, and 300 whole-station bootstrap resamples. Skill: RMSE in $x = V/K$, compared with persistence ($\hat{x}_{t+1} = x_t$) and the training-period station mean.

Writing note: Explain why r, d are fitted only at oases: there $\overline{W} \approx 1$, so growth and decay can be separated from water. If you fit free r, d, K on a planted section, only $r - d$ and r/K are identifiable.

4 Algorithms

Model 1 uses four algorithms: prepare the data (Alg. 1), fit r, d (Alg. 2), step the coupled system through one year (Alg. 3), and find the long-run state (Alg. 4). Thresholds and settings are listed in Section 5; the algorithms show only the logic.

Algorithm 1 Data preparation

Input: NDVI time series with pixel quality flags, for each station
Output: relative cover $x_{s,t}$; training and hold-out pairs

1: Keep only reliable pixels; build yearly growing-season cover $V_{s,t}$ (missing years stay missing)
2: $K_s \leftarrow 1.15\, Q_{0.95}$ of station s's training-period cover ▷ Eq. (9)
3: $x_{s,t} \leftarrow V_{s,t}/K_s$
4: Split consecutive-year pairs $(x_{s,t}, x_{s,t+1})$ by target year into training and hold-out sets

Algorithm 2 Calibration of r, d: pooled linear least squares

Input: training pairs from Alg. 1
Output: $\hat{r}, \hat{d}$ and their uncertainty

1: **Linearise.** With $\overline{W} = 1$, Eq. (1) becomes $\underbrace{x_{t+1} - x_t}_{y} = r\ \underbrace{x_t(1 - x_t)}_{z_1} - d\ \underbrace{x_t}_{z_2}$
2: Stack all stations' pairs: $\boldsymbol{y}$ and $\boldsymbol{Z} = [z_1,\ -z_2]$
3: **Solve** $(\hat{r}, \hat{d})^\top = (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{y}$ ▷ unique; Eq. (13)
4: **Uncertainty:** refit on resampled *whole stations* (cluster bootstrap)
5: **Check:** one-step error on hold-out pairs vs persistence $\hat{x}_{t+1} = x_t$

Key point. Fixing K beforehand makes the model linear in (r, d), so the fit is an ordinary regression with one exact answer. r and d are told apart because growth $x(1 - x)$ peaks at medium cover while decay x keeps rising.

Algorithm 3 One model year: water storage drives vegetation

Input: cover V_t, storage M, irrigation I_t, one year of daily rain, ET_0, temperature
Output: V_{t+1}, M at year end, $\overline{W}_t$

1: Demand D_j from current cover and season ▷ Eq. (2); V_t held fixed
2: **for** each day j **do**
3: $M \leftarrow M +$ effective rain + effective irrigation
4: Spill $M - C$ if storage is over capacity ▷ drainage
5: Transpire $T_{a,j} = \min(M,\ K_s(M)\, D_j)$; $M \leftarrow M - T_{a,j}$ ▷ stress when storage is low
6: $\overline{W}_t \leftarrow \sum_j T_{a,j} / \sum_j D_j$ ▷ share of demand met
7: $V_{t+1} \leftarrow$ Eq. (1) with $\overline{W}_t$
8: **return** V_{t+1}, M (carried to next year), $\overline{W}_t$

Key point. Water reaches vegetation only through storage: it carries over between days and years, overflow is lost, and growth uses the *share of demand met*, not the amount applied. Cover feeds back by setting the demand. Every day conserves water exactly (Eq. (4)).

Mathematical tools. (i) An explicit daily *mass-balance difference equation* (a "bucket" or linear-reservoir model); (ii) *piecewise-linear operators*: $\max(\cdot, 0)$ for overflow, the FAO-56 stress ramp K_s, and $\min(\cdot)$ so plants never take more than is stored; (iii) a *ratio aggregation* $\overline{W} = \sum T_a / \sum D$ that passes the fast (daily) water state to the slow (yearly) logistic map. The two time scales are nested: V is frozen within the year, M is updated every day.

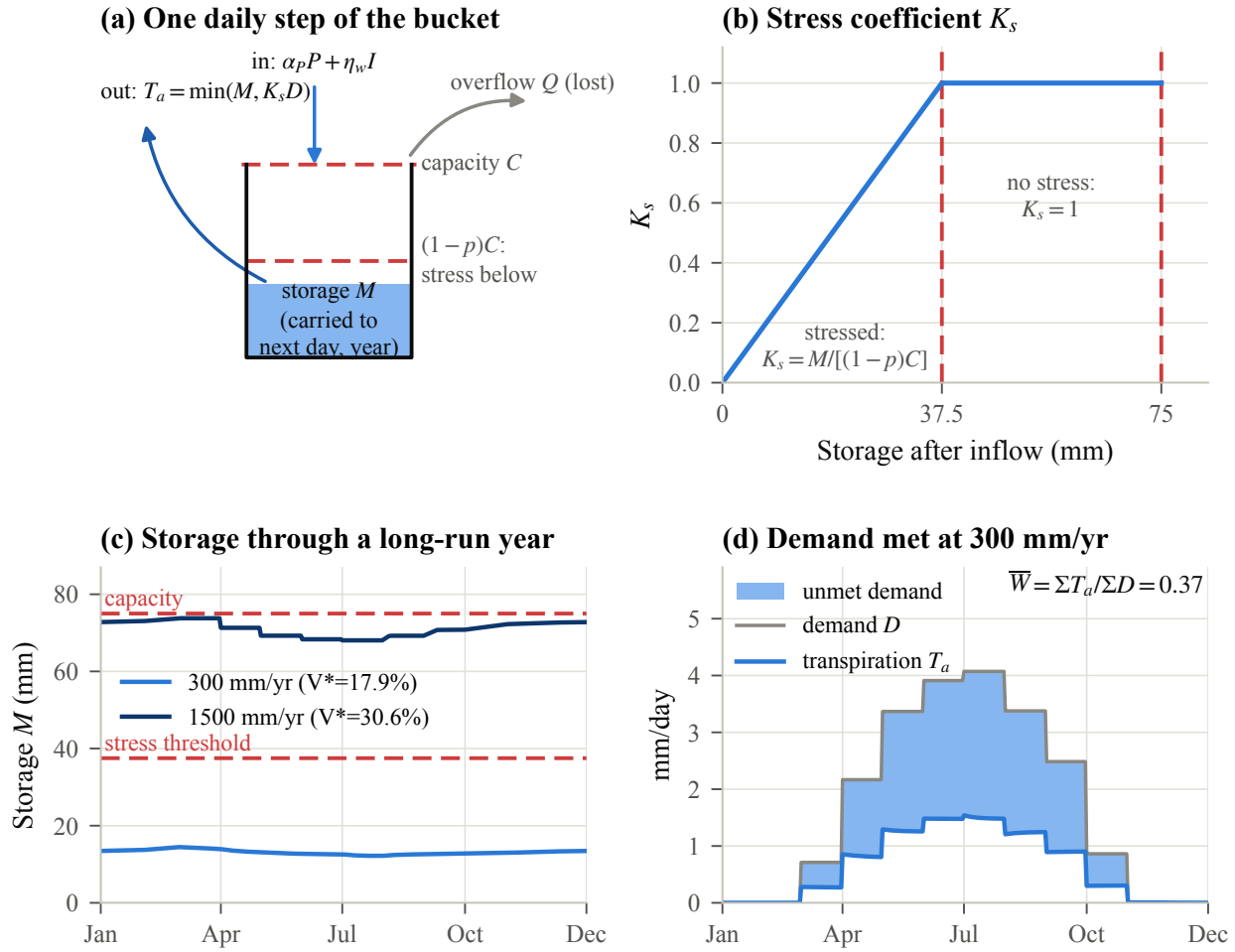


Figure 2: How Alg. 3 works. (a) One daily step. (b) The stress ramp: below $(1-p)C = 37.5$ mm transpiration is cut in proportion to storage. (c) Daily storage in a long-run year: at 300 mm/yr storage stays near 13 mm, so plants are stressed all year; at 1,500 mm/yr storage stays near capacity and 43.5 mm/yr overflows. (d) At 300 mm/yr only 37% of demand is met ($\overline{W} = 0.37$), which is what limits cover to 17.9%.

Algorithm 4 Long-run state for a given irrigation level

Input: irrigation I; a repeated typical climate year
Output: long-run $(V^*, M^*, \overline{W}^*)$

1: **for** a low and a high starting cover **do**
2: **repeat** apply Alg. 3
3: **until** V and M stop changing
4: Accept only if both starts reach the same point

Key point. Because $\overline{W}$ depends on V, the closed form Eq. (12) is only conditional; the long-run state is the fixed point of the coupled yearly map. Repeating Alg. 3 over many years from a planted start gives the scenario paths; varying one parameter gives the sensitivity table.
Code: Alg. 1–2: `Code/v7_model1.py` (`yearly_table`, `fit_stations`), `Code/v7_m1_run.py` (`bootstrap`); Alg. 3–4: `Code/v7_m1_water.py` (`water_year`, `equilibrium`, `trajectory`).

5 Parameters

Symbol	Value (range tested)	Unit	Meaning and status
r	0.547 (bootstrap 95%: 0.335–1.168)	yr^{-1}	intrinsic growth rate; fitted
d	0.107 (bootstrap 95%: 0.077–0.202)	yr^{-1}	natural decay rate; fitted
K	38.08 (section); per station $1.15Q_{0.95}$	%	cover ceiling; rule from data
C	75 (30–150)	mm	storage capacity; scenario
p	0.5 (0.3–0.7)	–	stress-free share of storage (FAO-56); scenario
c	1.0 (0.7–1.2)	–	transpiration coefficient; scenario
η_w	0.75 (0.65–0.85)	–	share of gross irrigation that reaches roots; scenario
α_P	0.25 (0–1)	–	share of rain that reaches roots; scenario
G	0	mm/yr	groundwater supply; scenario
V_{full}	66.59 (64.85–69.40)	%	full-cover end-member (local oasis NDVI 0.5827)
u_T	5	m/s	wind threshold in the index E (used in the output S only)
wind scale	1.0 (0.8–1.2)	–	multiplier on wind speed u_2 in ET_0; scenario
V_0, M_0	2, 37.5	%, mm	initial state of a new stand in scenarios

Table 2: Model 1 parameters. "Scenario" means no field data to fit it; values are stated assumptions with tested ranges.

Data and forcing.

- Stations: ALE, KEL, MF, QM, RQ, YL, YT (vegetated; all 7 pass the training missing-data rule). MODIS 250 m 16-day NDVI, pixel QA 0/1 only, ≥ 50% valid pixels per composite, ≥ 2 valid growing-season months per year. 395 of 396 growing-season station-months valid; no interpolated values are used for fitting or scoring.
- Climate for scenarios: 2015–2022 monthly climatology of Yutian/Minfeng (Open-Meteo archive): rain 52.2 mm/yr, ET_0 1,685.6 mm/yr, temperature-weighted ET_0 1,190.8 mm/yr; wind index peaks in April (2.29) and May (2.08).
- Raw MODIS replay: all 198 downloaded chunks match the original SHA-256 hashes.

6 Results

6.1 Calibration and model selection

Model	Parameters	Cond. number	AIC	RMSE train	RMSE hold-out	R^2 hold-out
(r, d) **adopted**	$r = 0.547 \pm 0.174$, $d = 0.107 \pm 0.033$	375	−286.48	0.0516	0.0635	0.463
(r, d) + wind term, tested and dropped	$r = 0.778$, $d = 0.105$, $k = 0.250 \pm 0.433$	4,406	−284.83	0.0514	0.0657	0.424
Persistence	$\hat{x}_{t+1} = x_t$	–	–	0.0572	**0.0536**	–
Station mean	training mean	–	–	–	0.0737	–

Table 3: Fits in units of $x = V/K$. Adding the wind-damage term worsens AIC and hold-out error and the uncertainty of k exceeds its value, so the term was removed (Section 3.4; all 18 tests in `wind_term_test.csv`).

- Hold-out skill by station (MSE skill vs persistence): MF +0.44, YT +0.20; ALE −0.44, RQ −0.52, YL −0.57, KEL −2.62, QM −2.88.
- Leave-one-station-out: r stays in 0.47–0.53 and d in 0.09–0.11 for six stations; leaving out YT gives $r = 0.958$, $d = 0.168$ (YT is influential).
- Bootstrap (300 resamples of whole stations, only 7 clusters): r median 0.554, 95% 0.335–1.168; d median 0.109, 95% 0.077–0.202.

- Stricter QA (75% valid pixels, 6 stations): $r = 0.510$, $d = 0.103$; conclusions unchanged.
- Free run from 2022 to 2025: the model stays near 2022 values while most stations greened in 2023–2025 (e.g. RQ predicted 33.0% vs observed 38.4% in 2025; QM 30.7% vs 37.2%).
- Section pixels (retrospective): station parameters transferred to planted pixels of CON1–3 give $R^2 = 0.398$ (RMSE 8.1 points, $n = 116$); CON4 gives $R^2 = 0.318$ (RMSE 12.1 points, $n = 129$). A free fit there gives net rate $r - d = 0.543$ and plateau 35.3%.
- Water bookkeeping: largest balance error over all scenarios 5.3×10^{-13} mm. No-water half-life: 6.1 years.

6.2 Irrigation–cover curve (long-run state)

Irrigation (mm/yr)	m^3/ha/yr	V^* (%)	M^* end of year (mm)	$\overline{W}^*$	Sand exposure $S^* = 1 - V^*/K$
0	0	1.77	7.1	0.21	0.95
50	500	6.05	8.3	0.23	0.84
100	1,000	9.43	9.4	0.26	0.75
150	1,500	12.17	10.5	0.29	0.68
200	2,000	14.43	11.5	0.32	0.62
300	3,000	17.94	13.4	0.37	0.53
400	4,000	20.54	15.4	0.43	0.46
500	5,000	22.54	17.3	0.48	0.41
600	6,000	24.14	19.3	0.54	0.37
800	8,000	26.51	23.2	0.65	0.30
1,000	10,000	28.20	27.0	0.76	0.26
1,500	15,000	30.60	72.8	1.00	0.20
2,000	20,000	30.60	74.4	1.00	0.20

Table 4: Long-run state for constant irrigation (fixed point, converged from both starts). The last column is the exposure index at average wind ($E = 1$), not a measured erosion reduction.

- **Diminishing returns.** The first 100 mm/yr adds 7.7 points of cover (0.077 points per mm); 500→600 mm adds 1.6 points (0.016 per mm); 800→1,000 mm adds 1.7 points (0.008 per mm).
- **Saturation.** Above about 1,500 mm/yr, $\overline{W} = 1$, storage is full and the extra water drains; cover is capped at $K(1 - d/r) = 30.6\%$.
- **Rain alone** (52 mm/yr) holds only 1.8% cover.

6.3 Scenario paths 2026–2050

Start: $V_0 = 2\%$, $M_0 = 37.5$ mm; climate repeats the 2015–2022 average year (not a weather forecast).

Scenario	2030	2035	2036	2040	2050
No irrigation	2.19	1.99	1.97	1.89	1.80
100 mm/yr	6.25	7.99	8.20	8.78	9.30
300 mm/yr	9.90	15.50	16.00	17.16	17.86
600 mm/yr	9.90	21.53	22.31	23.69	24.13
1,000 mm/yr	9.90	25.15	26.43	27.99	28.20
600 mm/yr, stopped after 2035	9.90	21.53	19.46	12.74	5.09

Table 5: Cover proxy V (%) at the end of each year.

- **Establishment is growth-limited.** Every quota of 300 mm/yr or more gives the same 9.90% in 2030: young stands use little water, storage stays full and the surplus drains. Water above ~300 mm/yr is wasted in the first five years.
- **Stopping water undoes the work.** After irrigation stops, $\overline{W}$ falls to 0.03–0.07 and cover drops from 21.5% to 12.7% in 5 years and to 5.1% in 15 years.
- It takes about 15–25 years to get close to the long-run value (300 mm/yr: 17.86% in 2050 vs 17.94% long-run).

6.4 Sensitivity

Parameter	Low / high	V^* at 300 mm/yr		V^* at 600 mm/yr	
		low	high	low	high
Baseline	–	17.94		24.14	
Transpiration coefficient c	0.7 / 1.2	21.32	16.22	27.12	22.49
Irrigation efficiency η_w	0.65 / 0.85	16.66	19.06	22.89	25.20
Rain share α_P	0 / 1	17.40	19.38	23.89	24.85
Full-cover end-member V_{full}	64.85 / 69.40	17.68	18.33	23.91	24.50
Storage capacity C	30 / 150 mm	17.94	17.94	24.14	24.14
Stress share p	0.3 / 0.7	17.94	17.94	24.14	24.14
Wind speed (through ET_0)	×0.8 / ×1.2	18.76	17.24	24.89	23.48

Table 6: Long-run cover under one-at-a-time changes.

- The demand side (c) and the efficiency of water delivery (η_w) matter most; each moves V^* by 1–3 points.
- Storage capacity and p do not change the long-run cover (they change M only). Reason: in a repeating year without overflow, storage returns to the same value, so yearly transpiration equals yearly effective inflow and $\overline{W}$ depends only on inflow and demand. Capacity matters for the timing within the year and for the start-up phase.
- Wind speed acts through evaporative demand: ±20% wind changes ET_0 by −8.3%/+7.7% (growing-season weighted) and long-run cover at 300 mm/yr by +0.8/−0.7 points; the exposure index S^* moves from 0.53 to 0.51/0.55 (`wind_channel.csv`).

7 What we can claim and what we cannot

Can claim	Must not claim
A mass-conserving coupled V–M model, calibrated on real NDVI with stated assumptions.	That the water response is field-calibrated (no measured irrigation–storage–cover data).
Growth and decay rates with honest, wide uncertainty.	That the model beats persistence for one-year forecasts (it does not).
Irrigation has strongly diminishing returns; establishment needs little water; stopping water causes collapse within ~15 yr.	Measured ground cover: V is an NDVI proxy; 1 km width and 128 km length are a study scenario, not an engineering boundary.
Wind lowers cover through evaporative demand (about 0.7 points for +20% wind at 300 mm/yr); denser cover leaves less sand exposed.	That wind does not damage plants: the test covers irrigated oasis stations at a yearly step only. Erosion reduction in tonnes or percent; CON4 as an independent validation (pixels were selected using later greening).

8 Figures ready for the paper

File (`Figures/M1/`)	What it shows / caption point
`m1_f01_station_series`	7 stations 2015–2025: observed V, one-step prediction, free run from 2022, persistence, local K
`m1_f02_holdout_skill`	Observed vs predicted; per-station hold-out RMSE vs persistence (negative skill kept)
`m1_f03_parameter_bootstrap`	r–d cloud from bootstrap, leave-one-out points, pooled fit
`m1_f04_cohort_diagnostic`	Planted pixels CON1–4 vs transferred curve (retrospective only)
`m1_f05_forcing_climatology`	Monthly rain, ET_0, temperature-weighted ET_0, wind index
`m1_f06_water_cover_curve`	Irrigation vs long-run V, $\overline{W}$, M, sand exposure S
`m1_f07_scenario_trajectories`	2026–2050 paths for constant quotas and the stop-after-10 case
`m1_f08_monthly_water_balance`	Monthly inflow, transpiration, drainage and storage
`m1_f09_sensitivity`	Tornado chart of water parameters; cover against wind speed (through ET_0)

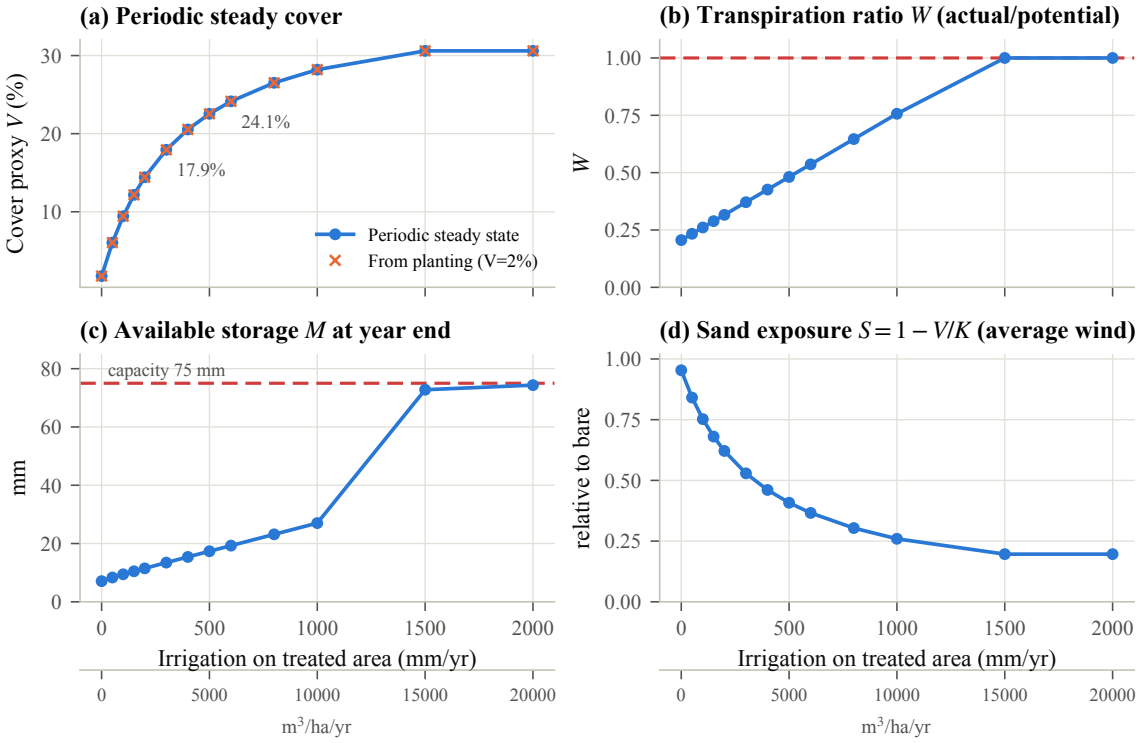


Figure 3: (From `m1_f06`) Irrigation–cover curve. Use it for the "diminishing returns" and "saturation" points.

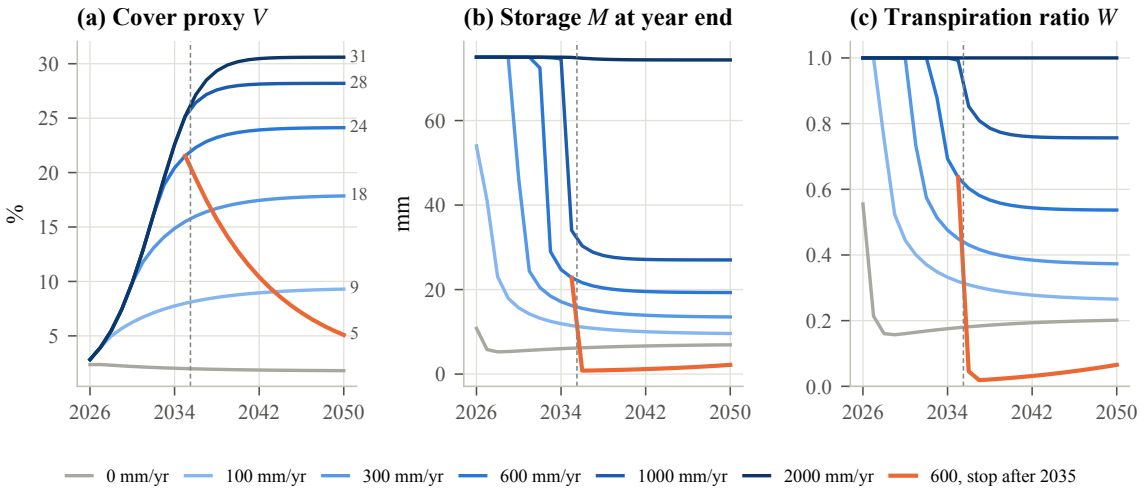


Figure 4: (From `m1_f07`) Scenario paths 2026–2050, including stopping irrigation after 10 years.

9 Suggested outline for the Model 1 section

1. Section choice and data (Yutian–Minfeng, 127.95 km × 1 km scenario; stations; QA rules).
2. Model structure (Fig. 1) and the four feedback links.
3. Equations (1)–(5) with one line each on meaning.
4. Algorithms: data preparation (Alg. 1), calibration as linear regression (Alg. 2), one model year (Alg. 3), long-run state (Alg. 4).
5. Parameters (Table 2); calibration result Table 3, honest skill statement, bootstrap.
6. Results: irrigation–cover curve (Table 4), scenario paths (Table 5).
7. Influencing factors: sensitivity (Table 6), wind through evaporative demand, sand exposure.
8. Limits and the hand-over to Model 2: each planted cohort carries its own (V, M).